Lab I3 - Camera in Isaac Sim: steer towards a red cube by its colour

Aim. Add an RGB + depth camera to the Isaac Sim robot and run a small ROS 2 node that finds a red cube in the image, turns the robot towards it, drives up to it and stops 0.55 m in front, using the depth image to measure the distance. A green cube is a decoy.

Time. 45 min. **Needs.** Only this folder, an **NVIDIA RTX GPU** (tested: RTX 5050 Laptop, 8 GB, driver 592.82) and about 15 GB of disk for Isaac Sim, which `pixi run isaac-install` downloads once for all Isaac labs on the machine. By running the scripts you accept NVIDIA's Isaac Sim licence.

This folder is a complete project

Everything this lab needs is in this folder. Nothing from the other labs is required.

Requirements: Windows 10/11, PowerShell, pixi (no administrator rights needed; install steps are in README-SETUP.md), and an NVIDIA RTX GPU with a current driver; about 15 GB disk for Isaac Sim (shared by all Isaac labs, installed once into `C:\ix`); internet for the first install.

In this folder	
`isaac\`	the Isaac Sim scene scripts and ROS 2 nodes
`scripts\`	helper scripts and the automated tests
`pixi.toml`	the environment and every `pixi run ...` command; `README.md` this manual; `tools\replay.ps1` re-runs the manual's commands as a self-test

Set it up once. `pixi install` puts ROS 2 Jazzy into `C:\rlabs` (a few GB; pixi's package cache makes the second lab on the same machine fast). Open PowerShell in this folder and run:

```
pixi install
pixi run verify
```

Run one lab at a time, and everything from this folder.

Theory

- Camera on the front of the body, 320 x 240 pixels, 90 degree field of view. Topics: `/camera/image_raw` (`sensor_msgs/Image`, `rgb8`: 3 bytes per pixel) and `/camera/depth` (`32FC1`: one float, metres, per pixel).
- The ROS 2 node `isaac\ros_nodes\color_follower.py` (about 60 lines, numpy only) does, per image: 1. a pixel is **red** if `R > 120`, `R - G > 70`, `R - B > 70`; 2. `error = (mean column of red pixels - centre) / half-width`, in -1 ... +1 (negative = target on the left); 3. turn `w = -1.2 * error`; drive forward only when the target is within 12 % of the centre; 4. stop when the depth in the middle of the picture is 0.55 m or less; with no red in view, turn slowly to search.
- Scene: the red cube is at (2.2, 1.3) - about 31 degrees to the left of the robot's nose - and the green decoy at (2.2, -1.2).

Procedure

1. T1 - Isaac Sim with the camera (window: robot, red cube, green cube).

```
pixi run isaac-p3 --gui
```

2. T2 - the images exist. Encoding, size and rate (the rate is in simulated time).

```
pixi run isaac-ros topic list --no-daemon --spin-time 5
pixi run isaac-ros topic echo /camera/image_raw --no-daemon --once --spin-time 5 --field encoding
pixi run isaac-ros topic echo /camera/depth --no-daemon --once --spin-time 5 --field encoding
pixi run isaac-ros topic echo /camera/image_raw --no-daemon --once --spin-time 5 --field width
```

3. T2 - start the controller. The robot turns left until the red cube is in the middle, then drives to it and stops. The terminal prints what the node sees.

```
pixi run isaac-follower
```

4. T3 - did it work? Wait about 40 s, then read the final position. The cube is at (2.2, 1.3); the robot should stop 0.55 m + half its length in front of it.

```
Start-Sleep 45
pixi run isaac-ros topic echo /odom --no-daemon --once --spin-time 5 --field pose.pose.position
```

Compute the distance from (x, y) to (2.2, 1.3). Recorded: 0.77 m (the node stops when the camera is 0.55 m from the cube's front face).

```
# Ctrl+C in T1 and T2
```

Observations (recorded)

Quantity	Value
Images	320 x 240, `rgb8` and `32FC1`
Start	red cube 31 degrees to the left, green decoy on the right
End	robot 0.77 m from the cube centre, 1 degree off the nose; it did not go for the decoy
Depth straight ahead (cube at 0.50 m by geometry)	0.48 m

Verification (own headless Isaac Sim, 4-6 min; close T1/T2 first)

```
pixi run test-isaac-p3
```

12 checks: image sizes and encodings; the red pixels are in the left part of the first picture; the depth value matches the geometry; the robot ends nearer to the red cube than to the green decoy, close to it and facing it.

Questions

1. Why is `R > 120` alone not a good "red" test? What does `R - G > 70` and `R - B > 70` add?

2. What would the node do if the red cube were behind the robot? If there were two red cubes?
3. Why does the node use *depth* to stop instead of the size of the red blob? Which one would a cheap robot with only an RGB camera have to use?
4. Change the colour test to follow the green cube instead. Which constants change?
5. *(Extension)* Add a lidar (see the LiDAR lab) and approach the cube without hitting a box.

If it does not work

Symptom	Fix
Robot only spins	the cube is not in view; check the camera picture in the Isaac Sim window; the colour test needs the light the script adds
`isaac-follower` prints nothing	T1 is not at `project 3 ready`
Camera images black	run with `--gui` once to let the shaders compile